

Jemmali Mohamed Mayssen

Software Engineering Student | Seeking a 6-Month End-of-Studies (PFE) Internship

📞 (+216) 27 429 398 | ✉ MohamedMayssen.Jemmali@esprit.tn | [🌐 linkedin](#) | [🐙 github](#)

Education

ESPRIT – Private Higher School of Engineering and Technology	2024 – Present
Software Engineering Degree (Master's Equivalent)	Tunis, Tunisia
Faculty of Sciences of Tunis	2021 – 2024
Bachelor of Science in Software Engineering & Information Systems	Tunis, Tunisia

Professional Experience

BeeCoders	June 2026 – Aug. 2026
AI & Full-Stack Engineering Intern (SmartPFE)	Lac 1, Tunis, Tunisia

- Designed and developed **SmartPFE**, an end-to-end AI platform guiding students across the graduation thesis lifecycle.
- Architected a **Corrective RAG (CRAG)** pipeline via **MongoDB Atlas Vector Search** (3,092 chunks, 384-dim multilingual embeddings), achieving **99.4% faithfulness** and **82.5% context recall**.
- Implemented an autonomous evaluation and query-rewriting loop with scoped semantic chunking, slashing prompt overhead by **70%** with an **80% composite quality score**.
- Stack** : NestJS, React, TypeScript, Python, Vector Search, Sentence Transformers, Ragas.

QantumShift	July 2025 – Aug. 2025
Machine Learning & Full-Stack Engineering Intern	Tunis, Tunisia

- Architected a decoupled 3-tier e-commerce platform: **Angular SPA**, **Spring Boot** REST API secured via **JWT**, and **MongoDB** persistence.
- Engineered a **collaborative filtering** recommendation engine: matrix factorization (**NMF**) on weighted implicit feedback (views vs. favorites) and **cosine similarity**.
- Resolved the **cold-start** problem via a popularity-based **fallback strategy** for new user discovery.
- Stack** : Angular, Spring Boot, Python (scikit-learn, pandas, numpy), MongoDB, JWT, REST API.

TUNISAIR	Jan. 2024 – June 2024
Full-Stack Developer Intern (Bachelor's Graduation Project)	Tunisair HQ, Tunisia

- Engineered a centralized B2B partner portal for travel agencies (contract fleet management, claims resolution, and financial reporting).
- Integrated the **Amadeus API (GDS)** for real-time flight synchronization and automated transactional credit line workflows.
- Secured architecture via **JWT** and enforced **RBAC** access control (front/back-office) with comprehensive audit logging.
- Stack** : Symfony, PHP, MySQL, JavaScript, Amadeus API, Bootstrap, SSMS, Git.

Academic Projects

DeepSkyn – AI Dermatology Companion 1 st Place Award ESPRIT Project Fair	Jan. 2026 – May 2026
--	----------------------

- Designed an AI-powered visual skin analysis platform with automated personalized skincare routine generation.
- Developed a dual ML pipeline combining **classification** (skin type, 82.3%) and **regression** (skin score, R^2 : 97.7%, MAE: 2.9) via **Random Forest** and **Gradient Boosting**.
- Implemented advanced security mechanisms: Google OAuth, IP/geolocation detection, and **device fingerprinting**.
- Stack** : React, NestJS, Prisma ORM, PostgreSQL, Python (scikit-learn), Docker, Git, CI/CD.

Technical Skills

Languages & Frameworks : Java, Python, TypeScript/JavaScript, PHP, C, Angular, React, Spring Boot, NestJS, Symfony.
Databases & DevOps : PostgreSQL, MySQL, SSMS, MongoDB, Neo4j, Docker, CI/CD, Git.
Data, AI & Tools : RAG, Vector Search, Scikit-learn, Pandas, NumPy, Postman, Figma, UML, Agile.